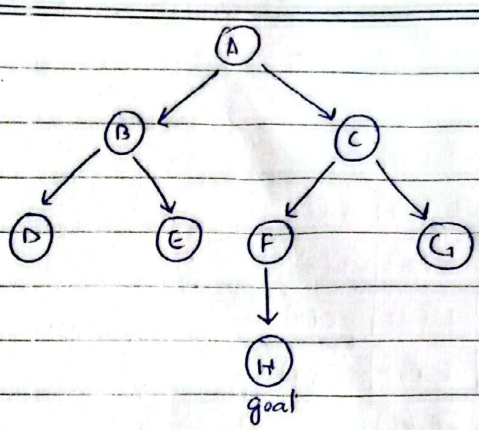
Liza
(15960)
Assignment # 01
(AI)

10

Date:

Question # 01

1	~~A~~			
2	~~B~~	C		
3	~~C~~	D	E	
4	~~D~~	E	F	G
5	~~E~~	F	G	
6	~~F~~	G		
7	~~G~~	H		
8	H			

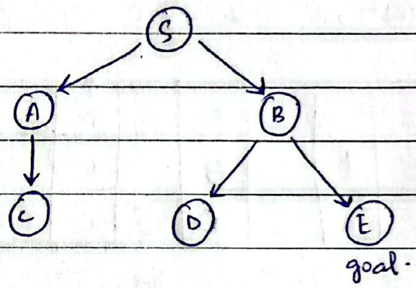


Final Path: A → C → F → H

Time Complexity: $2^3 = 8$

Question # 02

1	~~S~~		
2	~~A~~	B	
3	~~B~~	C	
4	~~C~~	D	E
5	~~D~~	E	
6	~~E~~		

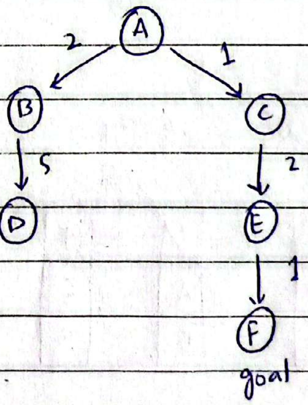


Final Path: S → B → E

Time Complexity: $2^2 = 4$

Question # 03

1	~~A(0)~~	
2	~~C(1)~~	B(2)
3	~~B(2)~~	E(3)
4	~~E(3)~~	D(7)
5	~~F(4)~~	D(7)



Final path: A → C → E → F

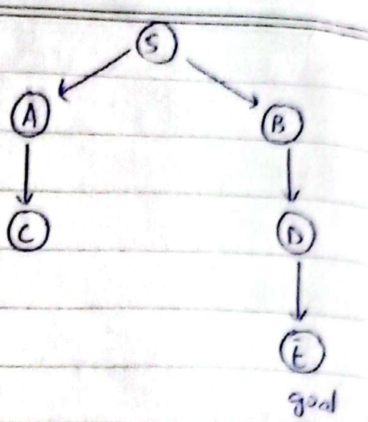
Total cost: 1 + 2 + 1 = 4

Time Complexity: $(2^4) = 16$

Date:

Question # 04

1	S (0)	
2	B (1)	A (3)
3	A (3)	D (3)
4	D (3)	C (7)
5	E (6)	C (7)
6	E (6)	

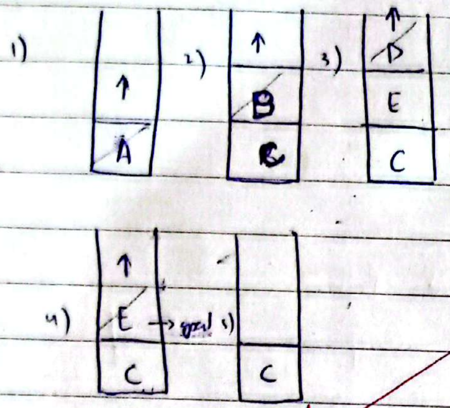


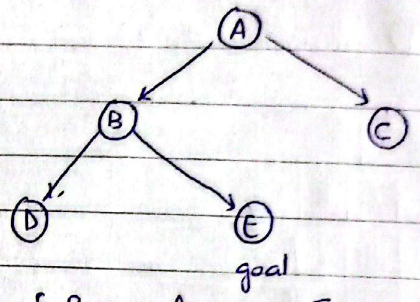
Final Path = S → B → D → E

Total Cost = greater than zero

T.C = $O(2^6) = 64$

Question # 05

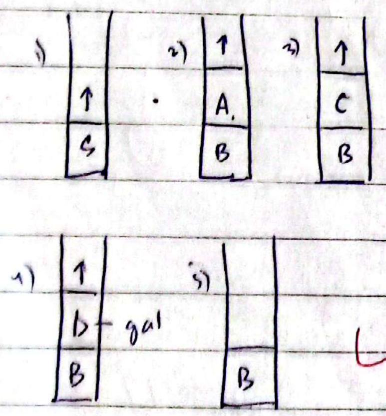


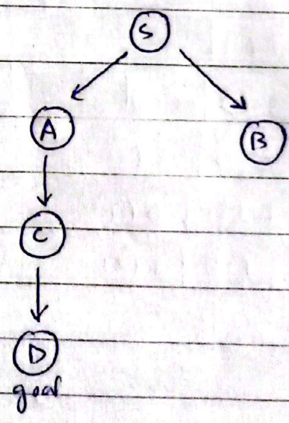


F.P = A → B → E

T.C = $2^2 = 4$

Question # 06



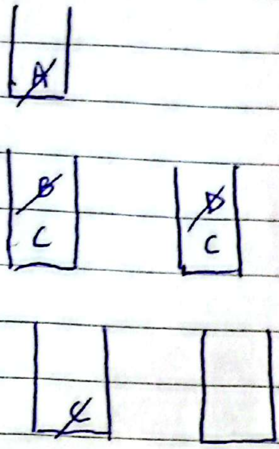


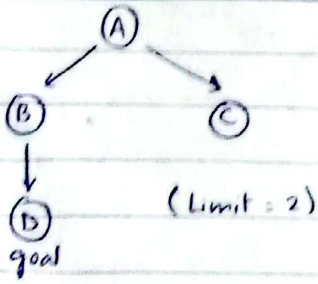
F.P = S → A → C → D

T.C = $2^3 = 8$

Date:

Question # 07

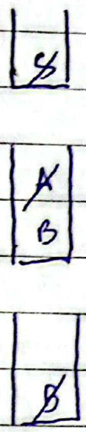


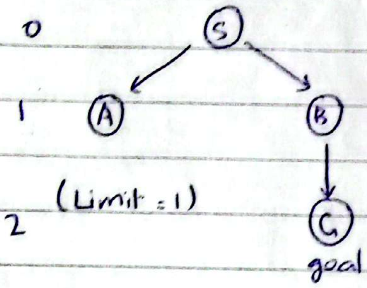


F.P = AB-D.

T.C = $O(2)^2 = 4$

Question # 08



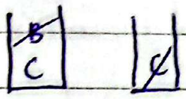


~~F.P~~ =
~~T.C~~

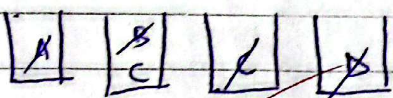
Question # 09

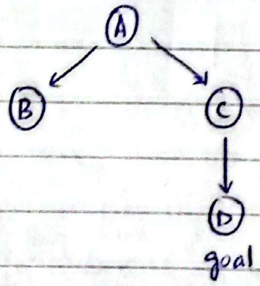
Iter 1, $\lambda = 0$

Iterat 2, $\lambda = 1$



Iterat 3, $\lambda = 2$



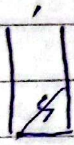


F.P = A-C-D

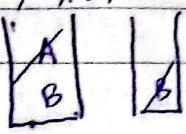
T.C = $O(2^2) = 4$.

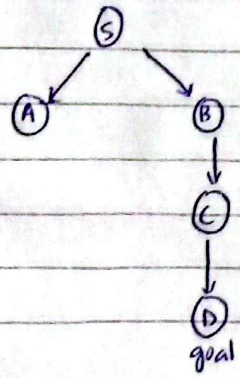
Question # 10

Iteration 1, $\lambda = 0$



Iteration 2, $\lambda = 1$



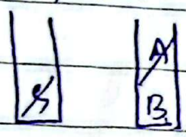


F.P

T.C

Date:

Iteration 3 , $\lambda = 2$.



R.P = S-B-C-D

T.C = $O(2^3) = 8$